Level: Bachelor
Programme: BE
Course: Probability and Statistics (New)

POKHARA UNIVERSITY

Semester: Fall

Year : 2025
Full Marks : 100
Pass Marks : 45
Time : 3 hrs.

Candidates are required to give their answers in their own words as far as practicable.
The figures in the margin indicate full marks.
Attempt all the questions.

1. a) A company monitors the download times in seconds of two software versions, version X and version Y, over 50 trials each. 7

Download time	5-7	7-9	9-11	11-13	13-15	15-17	17-18
Version X	5	10	12	10	8	3	2
Version Y	2	4	5	10	12	10	7

Which version has greater uniformity?

b) A tech company deploys updates to its software through three servers: Server X, Server Y and Server Z. Server X handle 50% of all updates, server Y handles 30% and server Z handles 20%. Historically, the probability that an update fails on server X is 2%, on server Y is 3% and on server Z is 5%. An update is randomly selected and found to have failed. What is the probability that this update came from server X, from server Y and from server Z? 8

2. a) A random variable X has the following probability function: 7

X	-2	-1	0	1	2
P(x)	0.2	0.1	0.3	0.3	0.1

i. $E(x)$
ii. $E(2x-3)$
iii. $V(2x-3)$

b) Out of 800 families in a certain city with 4 children each, what percentage would be expected to 8

i. 1 boy and 3 girl.
ii. at least one boy find the Expected number of families having 2 boys. Assume boys and girl having equal probabilities.

3. a) The mileage in thousands of miles which car owners get with a certain kind of type is a r.v having exponential distribution with 7

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$\lambda = 0.5$. What is the probability that such bulb will be still in operating condition after 4000 hours?

OR

In an examination 15% of the student got first class 60 marks or above, while 40% securing below 40 marks. Assuming the marks are normally distributed, estimate the mean and standard deviation.

b) An office switchboard receives telephone calls at the rate of 3 calls per minute on an average. If receiving of calls follows a Poisson distribution, find the probability of receiving: 8

i. no calls in one minute interval

ii. at least 3 call in a one minute interval

iii. at most 2 calls in 5 minute interval

4. a) Let the joint probability density function of two continuous random variables X and Y be given by 7

$$f(x,y) = f(x) = \begin{cases} k(x+y), & 0 < x < 2, 0 < y < 1 \\ 0, & elsewhere \end{cases}$$

i. Find the value of constant k.

ii. Obtain the marginal probability density function of X and Y.

iii. Find $P(X < 1, Y < 0.5)$

iv. Determine whether X and Y are independent or not.

b) A software company wants to estimate the average time in minutes required by its IT support team to resolve a specific type of software bug. A random sample of 8 bug-resolution time in minutes is recorded as follows: 8

42	38	45	40	44	39	41	43

Assuming that the bug-resolution times are approximately normally distributed, construct a 95% confidence interval for the population mean resolution time using t-distribution.

5. a) A population consists of the value 1, 2, 3, 6, 8. Prove that sample mean of size 3 is unbiased estimate of population mean (without replacement). 7

OR

Two independent random samples of 70 boys and 55 girls from normal population were drawn and found their mean weights were 62 kg and 46 kg with variances 16 and 12 respectively. Construct

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a 95% confidence for difference between mean weight of boys and girls.

b) Random sample of 400 male and 600 female were asked whether they would like to have a airport near their residence. 200 men and 325 women were in favor of the proposal. Test the hypothesis that proportion of men and women are in favors of proposal at 5% level of significance. 8

6. a) A dietitian wishes to see if a person's cholesterol level will change if the diet is supplemented by a certain mineral. Six candidates were pretested and then took the mineral supplement for a sis week period. The results are shown in the table. Can it be considered that the cholesterol level has been changed at the 0.1 level of significance? 8

Individuals	1	2	3	4	5	6
Before	210	235	208	190	172	244
After	190	170	210	188	173	228

b) General Manager of the New Laptop Shop of Tamrakar is concerned about the sales behavior of Lenovo brand laptops sold at the shop. He realizes that many factors might help the sales, among price major ones. Following data recorded. 7

y (Sales unit)	50	40	70	80	60	55
x (Price in '000 Rs)	70	50	80	90	100	45

i. Estimate the simple regression equation that best fits the data.

ii. Predict the value of sales when the price is Rs. 75000.

iii. Find coefficient of determination and interpret.

iv. Find correlation coefficient and interpret.

7. Write short notes on: **(Any two)** 2×5

a) Characteristics of good estimator

b) Error in hypothesis testing

c) Application of statistics in engineering

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